5 **(a)** *either* phase difference is π rad / 180°
or path difference (between waves from S_1 and S_2) is $\frac{1}{2}\lambda$ / $(n + \frac{1}{2})\lambda$. B1
either same amplitude / intensity at M
or ratio of amplitudes is 1.28 / ratio of intensities is 1.28^2 .... B1 [2]

(b) path difference between waves from S_1 and S_2 = 28 cm B1
wavelength changes from 33 cm to 8.25 cm B1
minimum when λ = (56 cm,) 18.7 cm, 11.2 cm, (8.0 cm) B1
so two minima .... B1 [4]

6 **(a)** **(i)** *E* = *V* / *d* C1